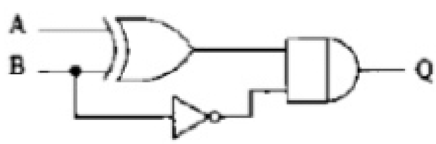
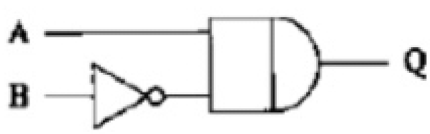
Srinidhi A
COMETFWC053

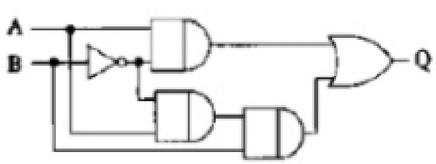
GATE Question No. 41

Question

For any set of inputs A and B, the following circuits give the same output Q, except one. Identify the circuit which gives a different output. (GATE PH 2010)

a)


b)


c)


d)


Question Analysis

From logical simplification of the circuits, the Boolean expression obtained is:

$$Q = \overline{A} + B$$

Each circuit is verified against this expression to determine the incorrect one.

The Truth Table

A	B	$Q = \overline{A} + B$
0	0	1
0	1	1
1	0	0
1	1	1

Hardware Implementation

The Boolean expression is implemented using the Arduino UNO board. Inputs A and B are provided through digital pins and the output Q is verified using an LED / 7-segment display.

Required Components & Pin Connections

Component	**Connection**
Input A	Digital Pin 2
Input B	Digital Pin 3
Output Q	Digital Pin 7
GND	GND
VCC	5V

Components Used:

- Arduino UNO
- Breadboard
- LED / 7 Segment Display
- Resistors
- Jumper Wires
- USB Cable

Logic Description

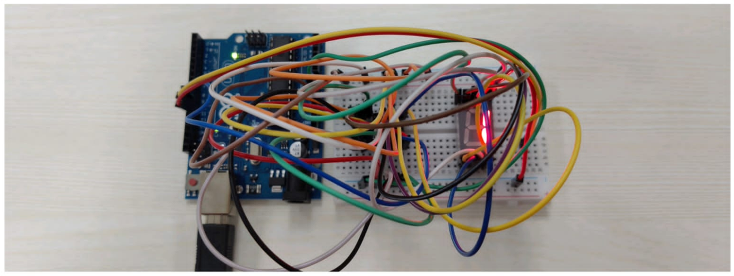
- Initialize inputs A and B.
- Implement the expression $Q = \overline{A} + B$.
- Apply all input combinations as per truth table.
- Verify hardware output with theoretical values.

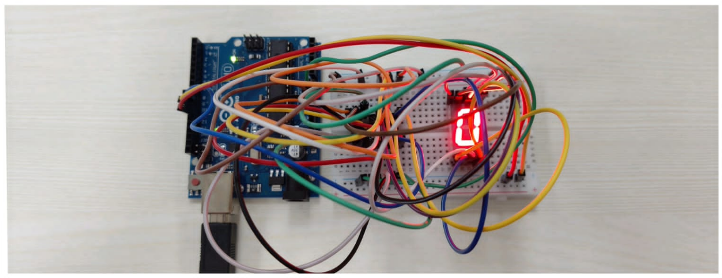
Code Uploading Steps

1. Create a Platform IO project.
2. Write the code in `main.cpp` inside the `src` folder.
3. Run the PIO project using the command `pio run`. This compiles the code and generates the .hex file.
4. Copy the .hex file to the Arduino Droid folder.
5. Connect the Arduino UNO to the mobile using an OTG cable.
6. Upload the hex file using the "Upload Precompiled" option.
7. Observe the output and verify the expression.

Experimental Truth Table

A	B	Observed Q
0	0	1
0	1	1
1	0	0
1	1	1





Conclusion

The Boolean expression $Q = \overline{A} + B$ was successfully implemented using the Arduino UNO board.

The hardware output matches the theoretical truth table. Hence, the experiment has been verified successfully.